FlowMLLab - Week 1.2

Pressure-Velocity Coupling in a Lid-Driven Cavity

SIMPLE, PISO and PIMPLE
A shared finite-volume Python implementation

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The central question

How does pressure enforce incompressibility, and how can different coupling loops reach the same flow with different computational costs?

Initial qualified case: Re=100. Two uniform grids. Executed fields, benchmark comparisons, momentum and continuity checks. An independent companion to the Week 1 streamfunction-vorticity formulation.

1. One physical problem, two formulations

Continuity and momentum: incompressible Navier–Stokes

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0$$

$$\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}=-\frac{\partial p}{\partial x}+\frac{1}{\mathrm{Re}}\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)$$

$$\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}=-\frac{\partial p}{\partial y}+\frac{1}{\mathrm{Re}}\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}\right)$$

Here u and v are horizontal and vertical velocity components, p is pressure divided by density times lid speed squared, x and y are positions divided by cavity side L, and t is time divided by L/U. Re = UL/nu, with lid speed U and kinematic viscosity nu. Momentum balances local acceleration and convection against pressure force and viscous diffusion. Continuity and both momentum equations must be solved together.

Use a unit square, unit lid speed and density, and viscosity 0.01. All walls are impermeable; tangential velocities are zero except for the moving lid.

Week 1 solves streamfunction and vorticity using the relations below. Pressure is recovered separately from steady momentum. Week 1.2 solves velocity and pressure together.

$$u=\frac{\partial \psi}{\partial y},\qquad v=-\frac{\partial \psi}{\partial x},\qquad \nabla^2\psi=-\omega$$

Why add this extension?

Students can inspect pressure as the continuity constraint, shared-face mass budgets, under-relaxation, and physical-time versus coupling iterations. The pressure-velocity formulation also provides a route toward three-dimensional solvers.

Learning outcomes: derive the pressure operator; distinguish the three loops; verify mass and gauge invariance; compare measured errors and cost; state the limits of a steady benchmark.

Pressure coupling: equations before algorithms

The incompressible Navier–Stokes system consists of continuity and two momentum equations. All quantities below are dimensionless, scaled by cavity side L, lid speed U and density rho. Pressure is divided by rho U squared; Re = UL/nu.

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0$$

$$\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}=-\frac{\partial p}{\partial x}+\frac{1}{\mathrm{Re}}\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)$$

$$\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}=-\frac{\partial p}{\partial y}+\frac{1}{\mathrm{Re}}\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}\right)$$

Here x and y are coordinates, t is physical time, u and v are velocity components, and p is pressure. Momentum balances local acceleration and convection against pressure force and viscous diffusion. At steady state the time derivative vanishes; convection remains. Continuity requires zero net flux.

Pressure is stored at cell centers, u on vertical faces and v on horizontal faces. The grid spacing is h = 1/N for N cells per side. Shared face fluxes cancel between neighbors.

$$(Dq)_{i,j}=\frac{u_{i+1/2,j}-u_{i-1/2,j}}{h}+\frac{v_{i,j+1/2}-v_{i,j-1/2}}{h}$$

Decode every matrix symbol

q collects the unknown interior face velocities of both components; it does not denote flow rate. A denotes their two momentum matrices. b collects wall and old-time sources. G maps cell pressure to face gradients; D maps face velocity to cell divergence.

$$Aq = b - Gp, \qquad R = A - \mathrm{diag}(A), \qquad d = \mathrm{diag}(1/A_{ii})$$

R is the off-diagonal neighbor-coupling matrix. d is an inverse diagonal operator, not grid spacing or a derivative. H is a computed source plus neighbor contribution, not an extra physical field.

$$H(q) = b - Rq, \qquad q = d(H - Gp)$$

Apply divergence to this velocity relation and impose continuity. This derives the pressure equation; its right-hand side is the mass imbalance of the pressure-free trial velocity.

$$0 = D(dH) - DdGp, \qquad L = -DdG, \qquad Lp = -D(dH)$$

Pressure gauge

Only pressure gradients affect velocity. Pin one pressure cell to solve the singular closed-cavity system, then remove the mean for reporting. A constant pressure shift leaves velocity unchanged.

SIMPLE: derive the mass correction

SIMPLE means Semi-Implicit Method for Pressure-Linked Equations. k counts steady coupling iterations, not physical time. Predict velocity using the current pressure and relaxed momentum system.

$$Aq^{*} = b - Gp^{k}$$

A star means tentative velocity. A prime denotes a correction, not differentiation. Approximate the velocity correction using the momentum inverse diagonal; then demand zero corrected divergence.

$$q' = -dGp', \qquad 0 = Dq^{*} - DdGp'$$

$$Lp' = -Dq^{*}, \qquad q^{k+1} = q^{*} - dGp'$$

$$p^{k+1} = p^{k} + \alpha_p p'$$

alpha_p is pressure relaxation. alpha_u is momentum equation relaxation: divide the original diagonal by alpha_u and add (1/alpha_u - 1) times the original diagonal times the previous velocity to the source. Both lie between zero and one. Reassemble convection after correcting velocity.

Why the full velocity correction?

Its pressure equation was derived to cancel the complete mass defect. Scaling the face correction by alpha_p leaves part of that defect. Relax pressure, but apply the conservative velocity correction fully. Check the original nonlinear momentum equations, not merely the size of a relaxed update.

PISO: fixed matrix, changing neighbor contribution

PISO means Pressure-Implicit with Splitting of Operators. n labels physical time; m labels inner corrections. Backward Euler adds 1/Delta t to the diagonal and q at the old time divided by Delta t to the source. Save the old-time velocity once.

$$\frac{q^{n+1}-q^{n}}{\Delta t}+\mathrm{C}(q^{n+1})=-Gp^{n+1}+\mathrm{V}(q^{n+1})$$

C denotes discretized convection; V denotes viscous diffusion. First freeze convection coefficients, assemble A and b, and solve one momentum predictor. Hold A, b, d and L fixed through all inner corrections.

$$H^{(m)}=b-Rq^{(m)}$$

$$Lp^{(m+1)}=-D(dH^{(m)})$$

$$q^{(m+1)}=d(H^{(m)}-Gp^{(m+1)})$$

This implementation solves absolute pressure in each correction: replace the pressure; do not add it to the previous pressure. Updated neighbor velocities change H at the next correction. Two sparse-solver iterations on an unchanged pressure system do not constitute two PISO corrections.

Scope of improvement

Inner corrections improve consistency with the frozen momentum matrix. They do not refresh nonlinear convection coefficients or remove first-order backward-Euler time error. Advance physical time once after all corrections.

PIMPLE: two loops, one physical time advance

The outer index labels momentum coefficient updates; the inner index labels pressure/velocity corrections. Save old-time q once. Each outer loop rebuilds convection using the latest corrected velocity, assembles a new backward-Euler momentum system and solves a predictor. Its inner loop then performs the PISO equations on that frozen matrix.

Follow one step

Save old-time velocity → outer loop: rebuild momentum and predict → inner loop: update H, solve pressure, reconstruct velocity → repeat inner corrections → repeat outer momentum loops → accept fields and advance time once.

N_o = outer_correctors and N_i = pressure_correctors. One accepted PIMPLE step uses N_o momentum assemblies, 2 N_o component momentum solves, and N_o N_i pressure solves. Three outer and two inner loops therefore require six pressure solves and six component momentum solves. Old-time sources remain unchanged throughout all loops.

Interpretation and cost

One outer loop reproduces this PISO implementation. Additional outer loops address lagged nonlinear coefficients; additional inner corrections address coupling for a fixed matrix. More loops cost more work and need not be economical for a steady cavity. A larger stable step does not imply an accurate transient trajectory.

A student checklist for a fair comparison

Separate accuracy, convergence and cost. Accuracy uses independent velocity reference data and mesh refinement. Convergence checks each method's equations and mass balance. Cost includes solver seconds, momentum solves and pressure solves. An iteration has a different meaning in SIMPLE, PISO, PIMPLE and streamfunction–vorticity.

Keep native residuals separately labeled: a vorticity update norm is not an FV momentum residual. Equal residual numbers across formulations do not ensure equal accuracy. Compare common centerline errors, final velocity fields and timings at stated resolution spacing and tolerances.

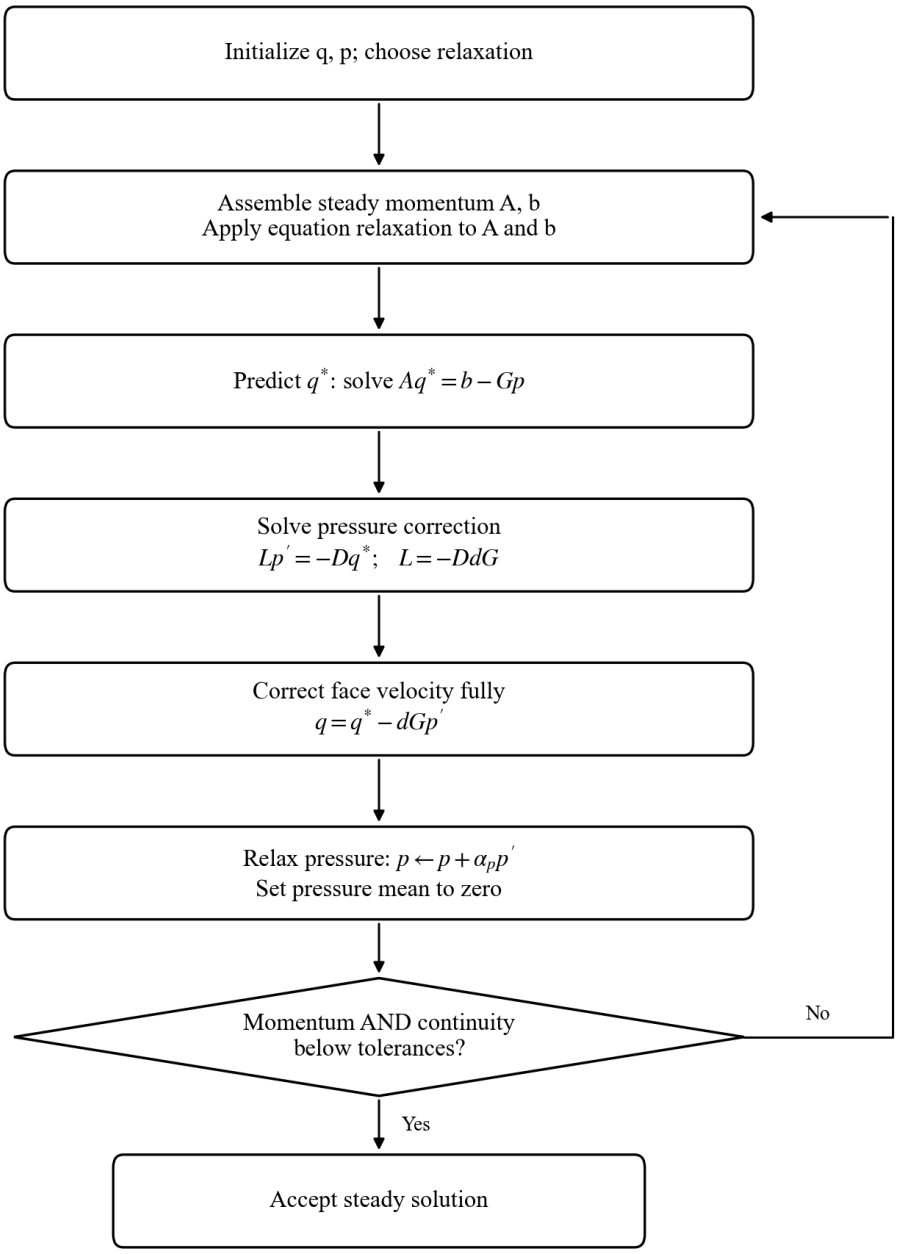
Time cases sequentially on the same machine. N FV cells correspond to N+1 streamfunction nodes. Higher Re and finer grids can require substantially more memory and runtime; begin with the coarse case before using a laptop or limited Colab runtime for refinement.

Questions to answer

Which quantities change in an inner correction? Which additional quantity changes in an outer loop? Why can divergence be tiny while momentum error remains large? Why must old-time velocity stay fixed inside both loops? Why is a steady benchmark insufficient to validate a transient time step?

SIMPLE: follow the control flow

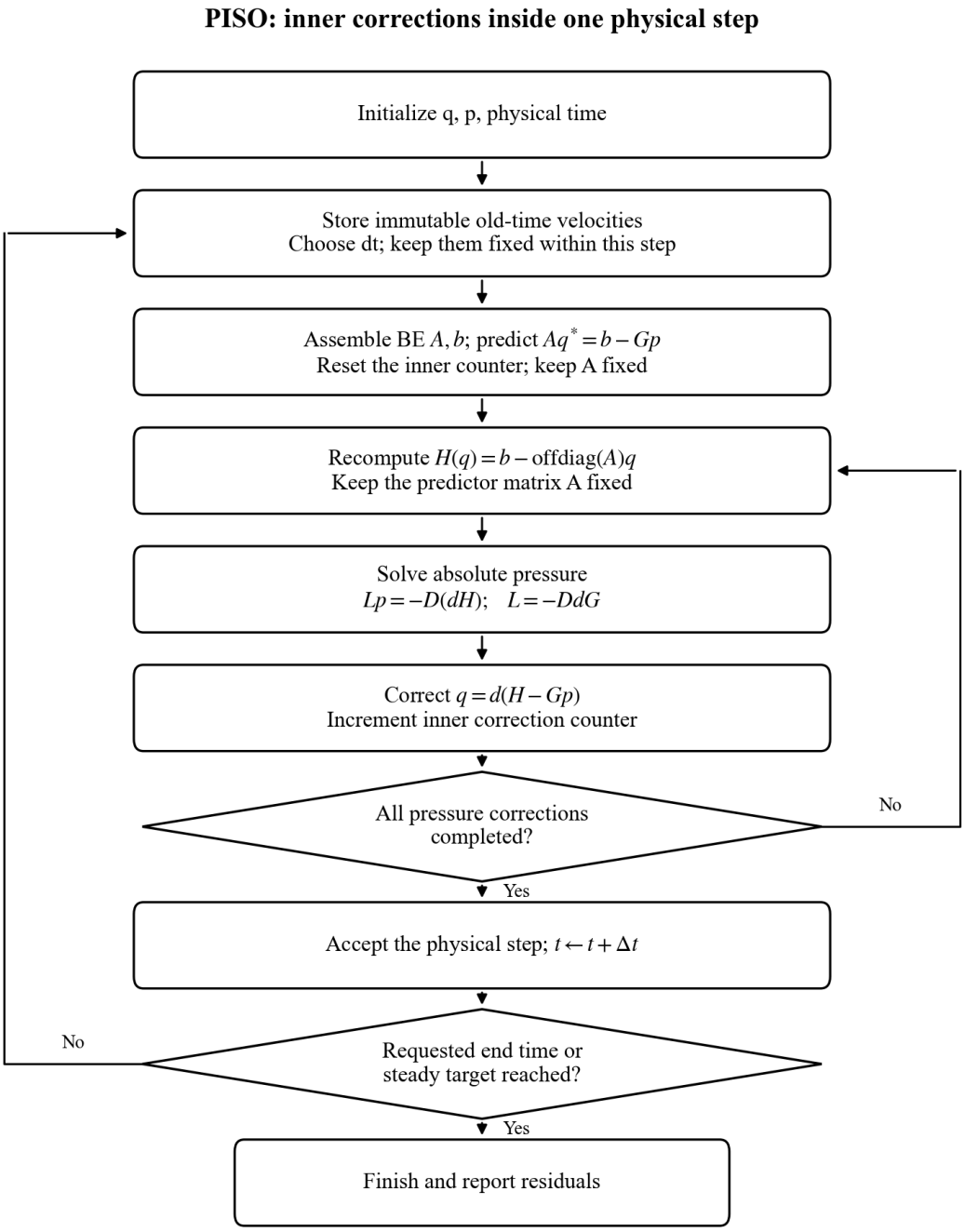
SIMPLE: a steady coupling iteration



The iteration index is not physical time. $d_i = 1/A_{ii}$.

Read the symbol definitions and loop indices in the preceding lesson before following the arrows. Old-time velocity remains fixed inside a physical step.

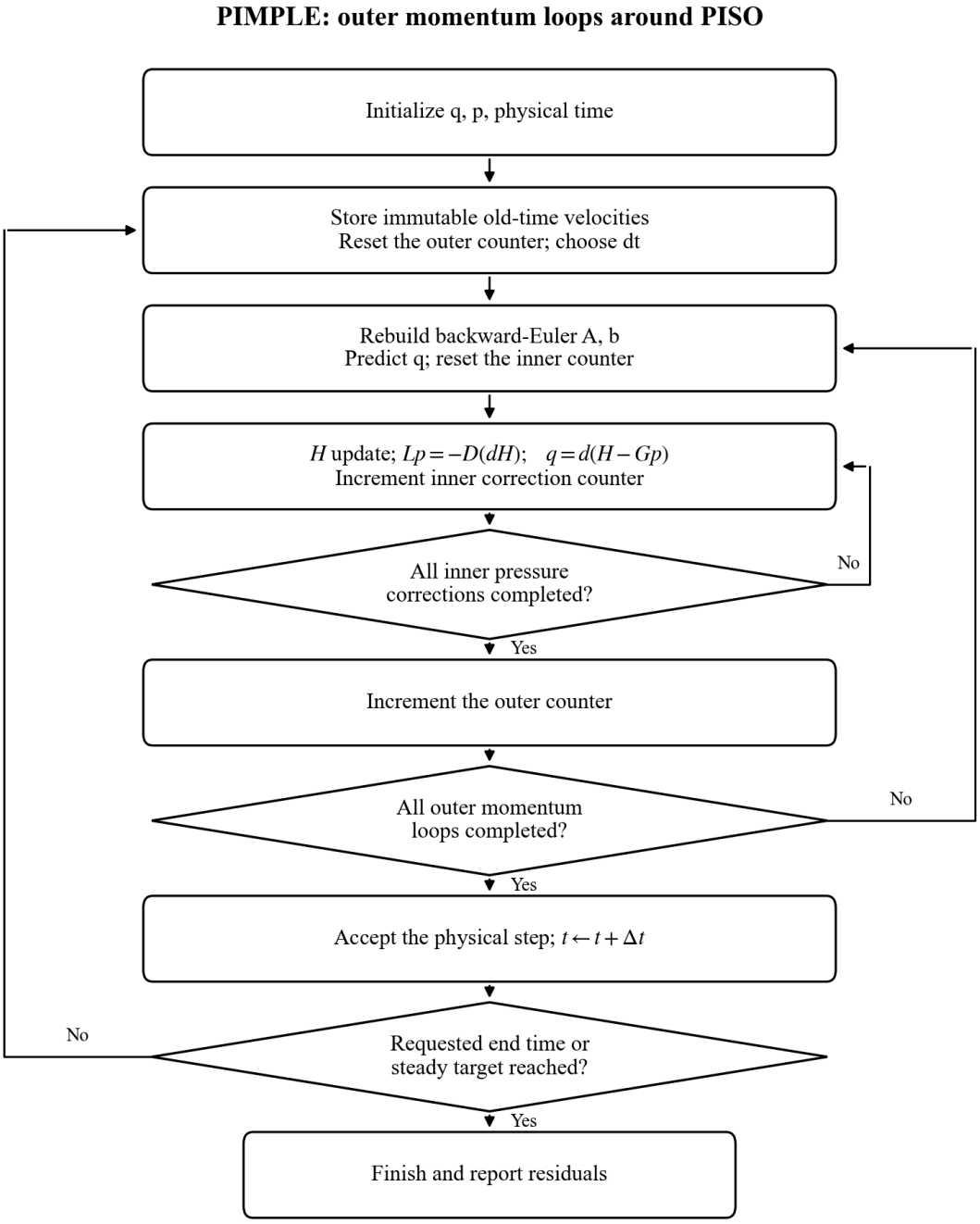
PISO: follow the control flow



An inner correction updates H; it is not a linear-solver sweep.

Read the symbol definitions and loop indices in the preceding lesson before following the arrows. Old-time velocity remains fixed inside a physical step.

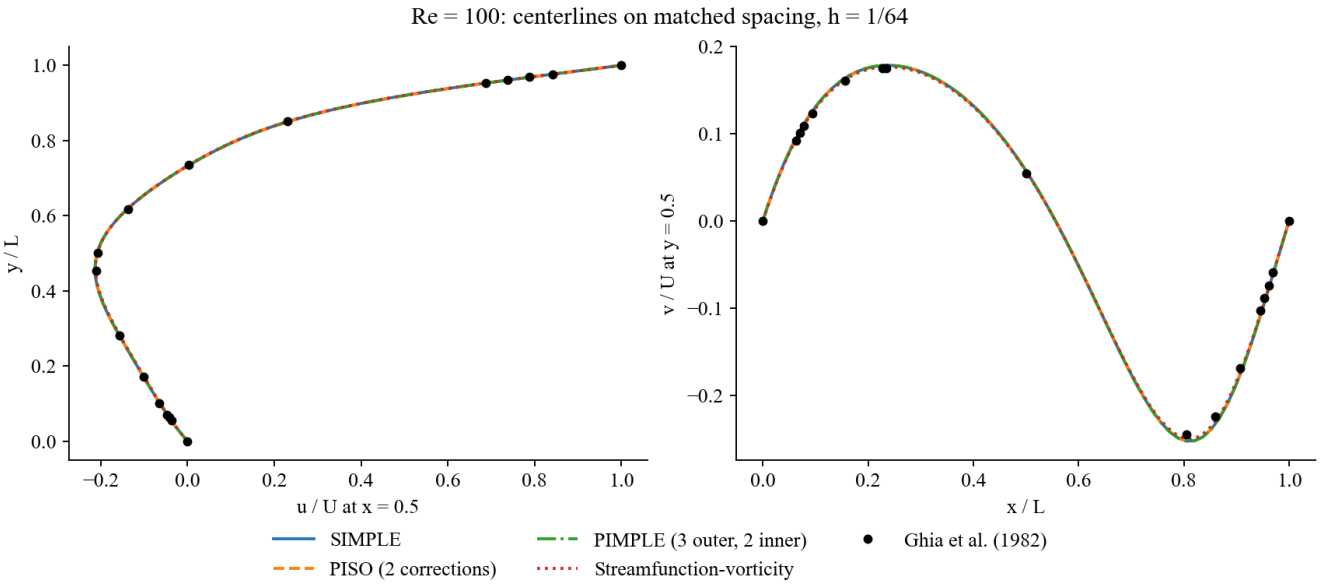
PIMPLE: follow the control flow



Old-time fields stay fixed in BOTH loops. One outer loop = PISO.

Read the symbol definitions and loop indices in the preceding lesson before following the arrows. Old-time velocity remains fixed inside a physical step.

6. Benchmark velocity and two-grid evidence



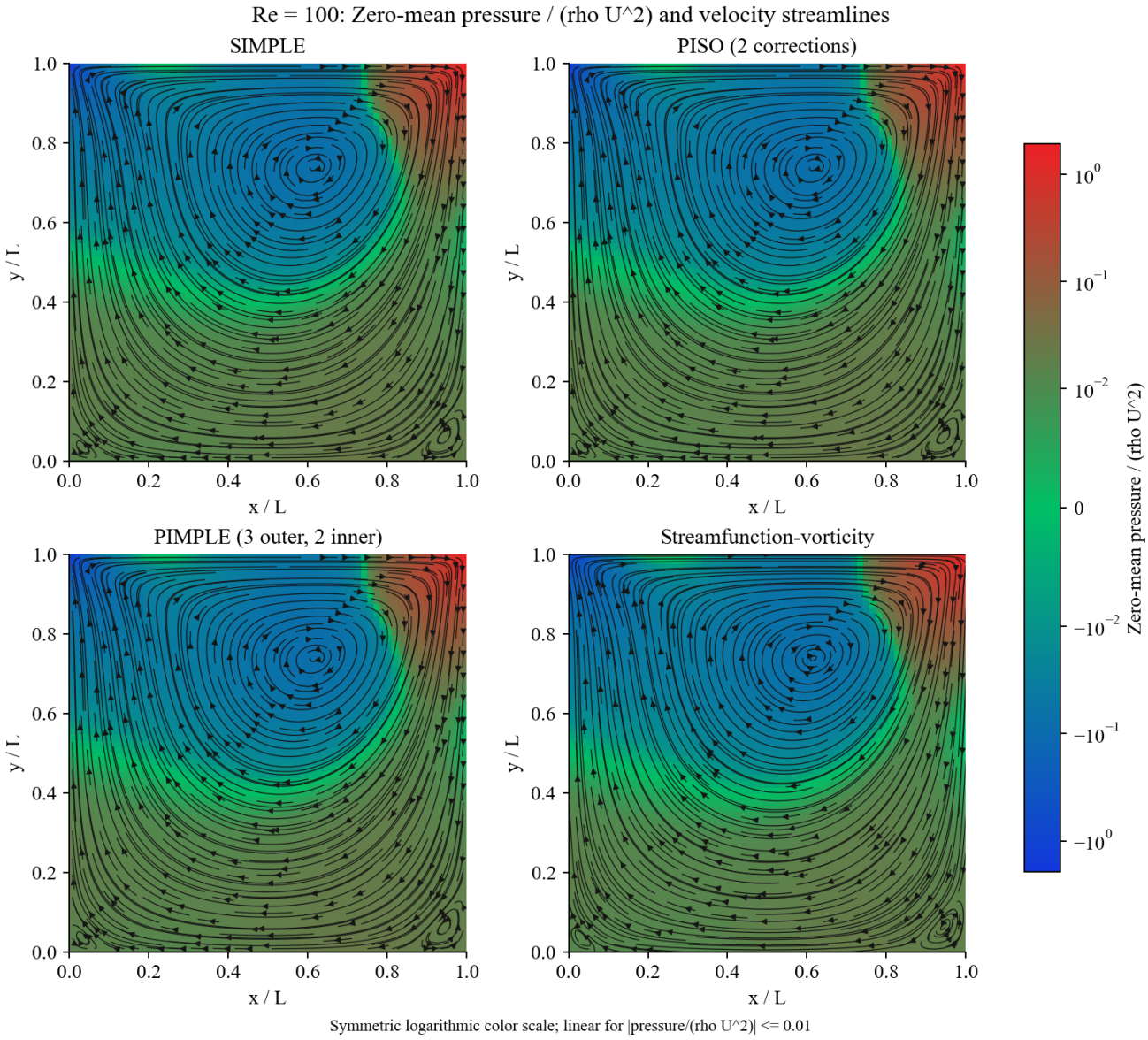
Ghia, Ghia and Shin (1982) velocity markers are compared at their stated coordinates, including wall endpoints. MAC face locations and FD nodes are interpolated only for the declared centerline diagnostics.

Fair resolution comparison

32/64 FV cells per side correspond to 33/65 FD nodes: compare h, not array length. The three FV algorithms use identical spatial equations. The independent FD formulation is a cross-check with a different truncation error.

The FD reference errors decrease on refinement; FV errors against Ghia increase slightly between these two grids. This nonmonotone trend is retained explicitly. Coarse-grid agreement with discrete benchmark data does not establish continuum accuracy. Further refinement and another reference are needed; no asymptotic order is claimed. Ghia velocity tables do not validate pressure.

7. Physical fields and pressure interpretation

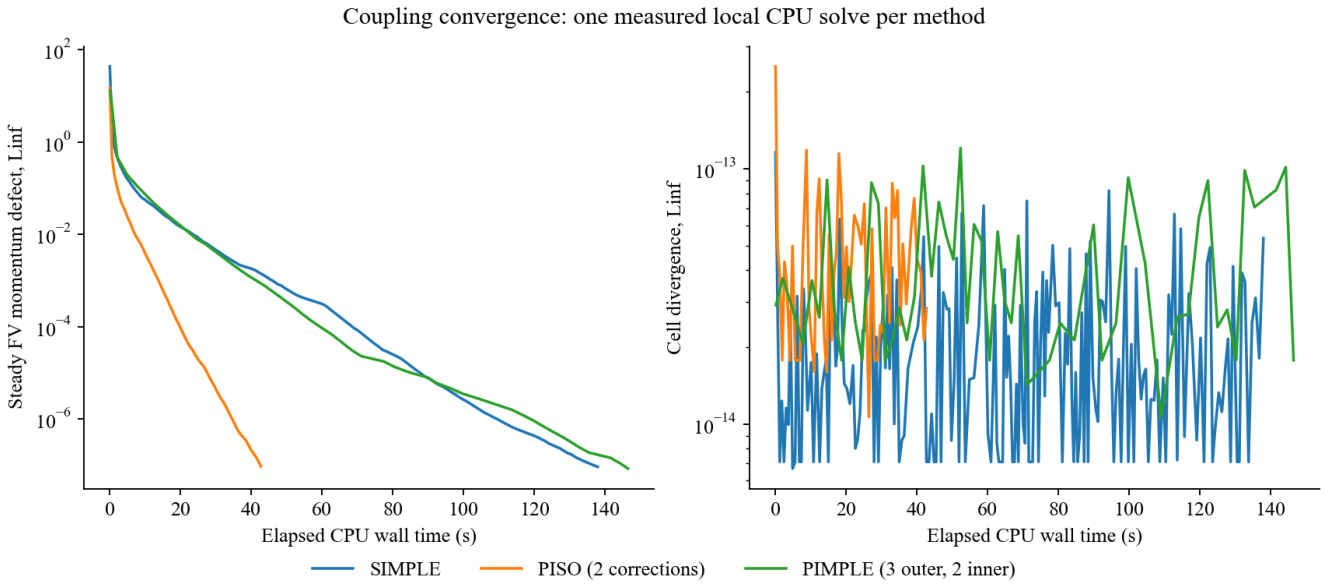


All pressure panels use the same zero-mean convention and common color limits. The three converged FV fields agree closely; the FD recovered pressure can differ near the singular lid corners.

Pressure evidence

Comparing pressure between independent formulations is informative but is not an exact-pressure validation. The notebook also shows speed, streamlines and vorticity on shared physical scales.

8. Convergence, cost and stopping conditions



The plotted steady FV momentum defect has units U^2/L, with $U=L=1$. Continuity is the maximum cell divergence. The elapsed-time axis includes actual numerical work, so differences in loop cost remain visible.

The FD reference uses a vorticity-update stopping norm and is not overlaid as if it were the same FV residual. A small under-relaxed update is not a substitute for a momentum-equation residual.

Measured conclusion

All six new FV runs reach the declared steady tolerance. On each grid, the three methods agree within 2×10^{-6} in u, v and zero-mean p. Timings are one local CPU solve per method, not a universal speed ranking.

9. What comparative studies actually show

Different studies answer different questions. None of these sources is a controlled comparison of our three Python implementations. There is no universal fastest coupling method.

Issa (1986): the transient motivation

PISO approximates the implicitly discretized equations through staged pressure/velocity corrections. Its companion study reports lower effort for time-evolving flows at sufficiently small splitting error. These publisher abstracts do not justify arbitrary large steps or universal steady superiority.

JCP 62, 40-65; companion paper, 66-82.

Pascau and Garcia (2010): ranking depends on the case

Their collocated-grid cavity study favored standard PISO among the tested variants at Re=1000. At Re=5000, a SIMPLEC-based variant performed better over a broader relaxation range. This is not a PIMPLE test and does not transfer as a MAC-grid ranking.

Enhancement of PISO Scheme in Collocated Grids, ECCOMAS CFD 2010.

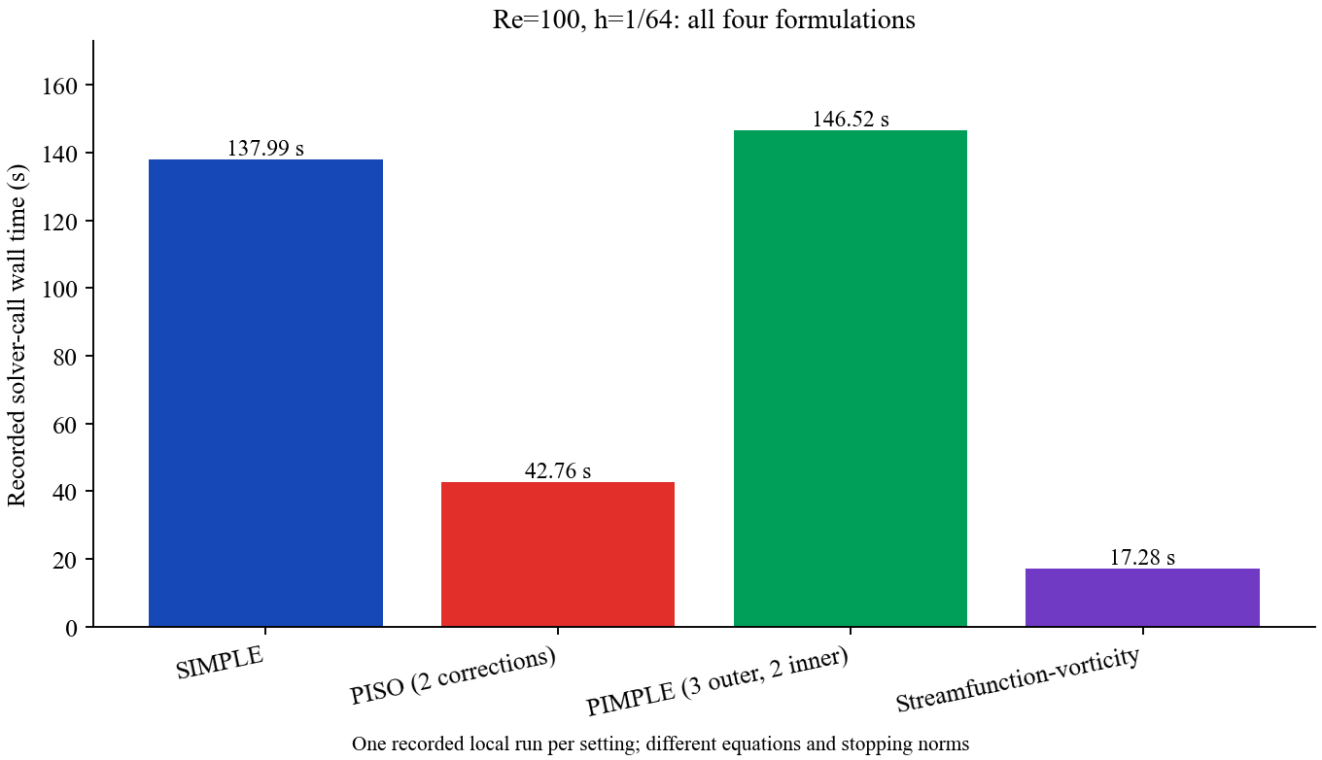
Brogi et al. (2024): coupling and time-step accuracy

A modern OpenFOAM precision study discusses PISO/PIMPLE time-step choices in a compressible shock tube. Extra coupling can make larger steps useful, but increasing dt still affects accuracy. This is not an incompressible-cavity speed comparison.

On floating point precision in CFD using OpenFOAM, FGCS 152, 1-16. The 2022 preprint is an earlier version with different settings.

Our assessment rule: compare measured cost at a specified field-error and equation-defect target; retain failures. An iteration count alone is insufficient.

Recorded Re=100 time: include the independent formulation



All four recorded methods appear in this timing figure. Streamfunction-vorticity has its own stopping criterion and reconstructs pressure afterward. The preceding FV convergence curves omit it because the vorticity-update norm is not a momentum residual; omission from that residual axis must not omit its measured runtime.

10. Higher Reynolds number: separate the causes

There is no universal Reynolds-number limit of SIMPLE, PISO or PIMPLE. Increasing Re can change spatial resolution requirements, nonlinear coupling and the physical/model behavior simultaneously.

A. Spatial convection and mesh resolution

$$a_E = \frac{\nu}{h^2} - \frac{u_E}{2h}, \qquad \mathrm{Pe}_h = \frac{|u_E| h}{\nu}$$

$$\mathrm{Pe}_h \le 2 \implies a_E \ge 0$$

The displayed bound is a sufficient positivity condition for this central neighbor coefficient, not a universal solver-failure threshold. Near unit advective speed, Re/N is about 15.6 for Re=1000 on 64 cells. Pressure corrections alone cannot resolve missing boundary-layer or vortex scales.

B. Nonlinear equation consistency

Reducing viscosity strengthens convection relative to diffusion. Relaxation, smaller physical steps or outer coefficient updates can improve discrete equation convergence. Measure these changes at the same spatial scheme and accuracy target.

C. Physical/model behavior

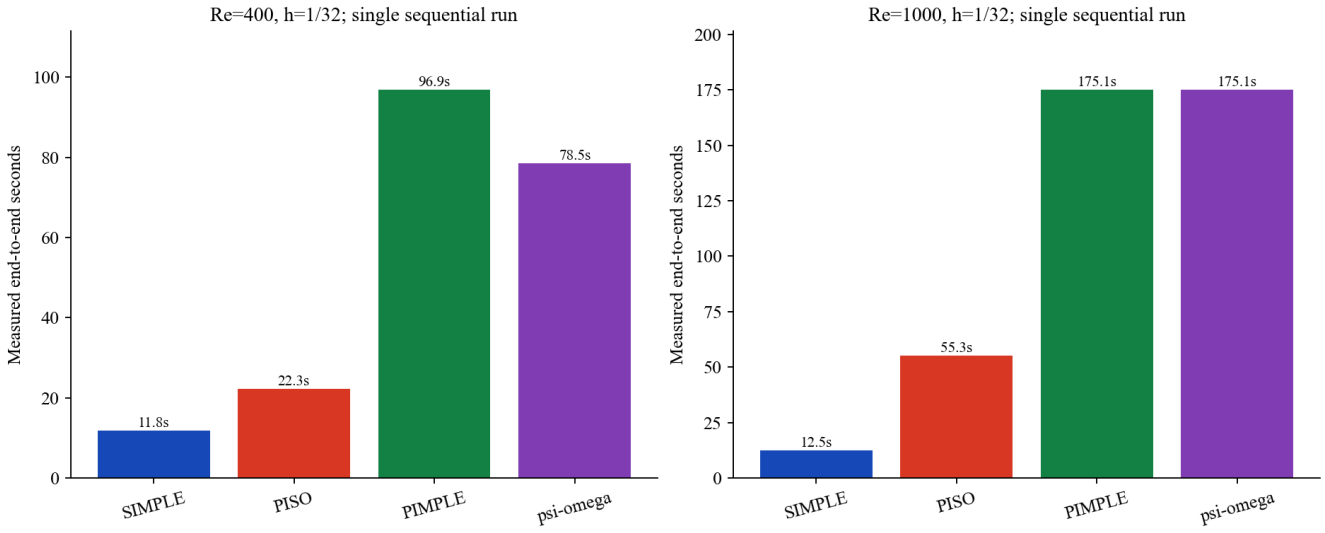
A real higher-Re cavity can develop unsteady or three-dimensional motion. A 2D steady solver does not establish that physical state. Nonconvergence alone does not prove an algorithm bug or the need for a turbulence model.

Albensoeder and Kuhlmann (2005), Accurate three-dimensional lid-driven cavity flow.

Implemented study boundary

The baseline CavityConfig remains restricted to Re=100. The separate experimental wrappers retain executed Re=400/1000 calculations, native steady-equation gates, independent velocity errors and resolution studies. They do not establish turbulent, three-dimensional or transient accuracy.

Executed higher-Re comparison: all four formulations

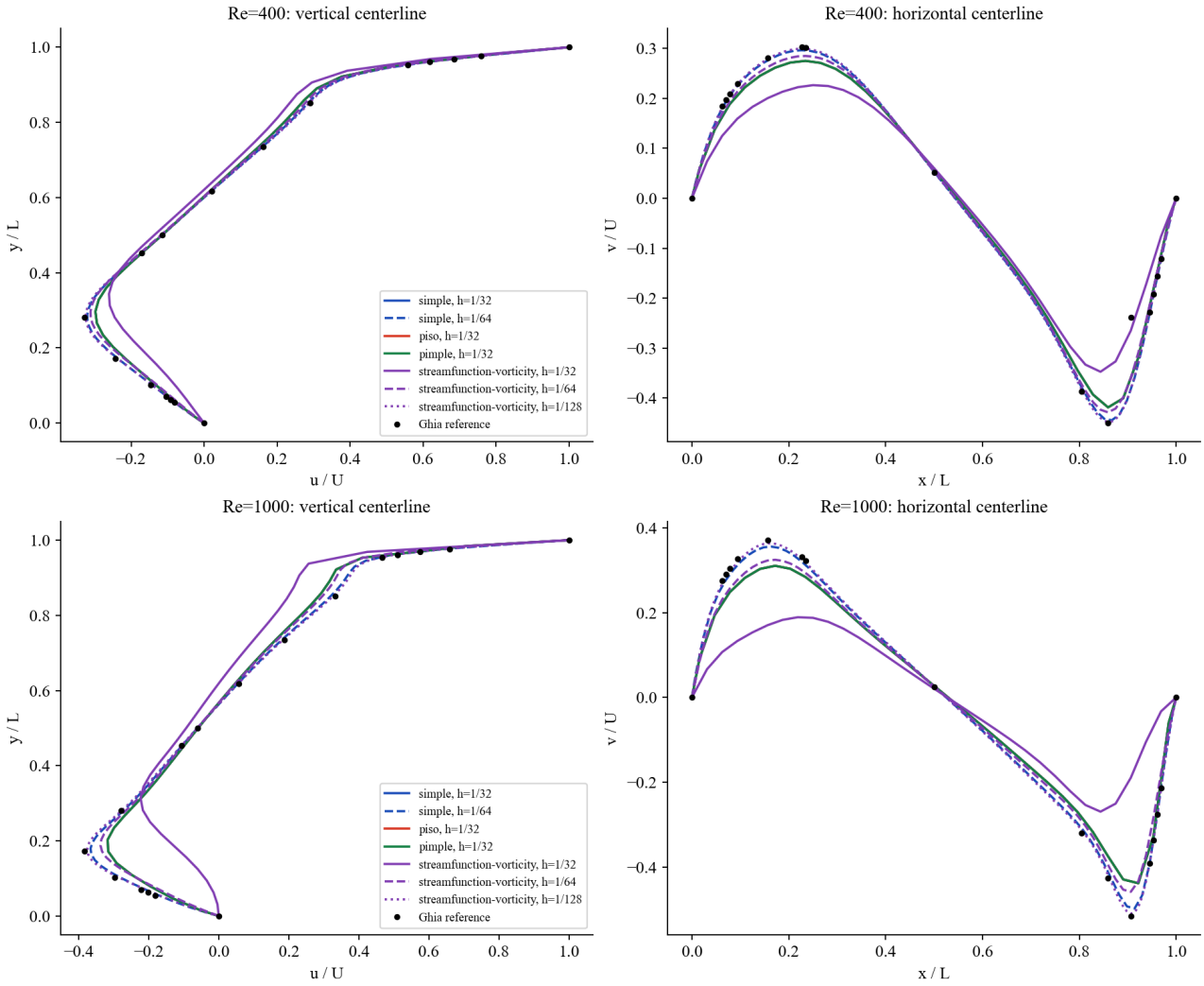


Re=400 and 1000, h=1/32. FD uses 33x33 nodes; FV uses 32x32 cells. These are sequential local solver-call timings, including FD pressure recovery, with different native stopping norms. Compare reference error before interpreting seconds. They are not a universal speed ranking.

Re	Method	Seconds	u error (%)	v error (%)
400	pimple	96.95	3.61	14.77
400	piso	22.28	3.61	14.77
400	simple	11.79	3.61	14.77
400	streamfunction-vorticity	78.47	11.89	26.73
1000	pimple	175.14	8.67	14.15
1000	piso	55.30	8.67	14.15
1000	simple	12.46	8.67	14.15
1000	streamfunction-vorticity	175.07	34.69	60.14

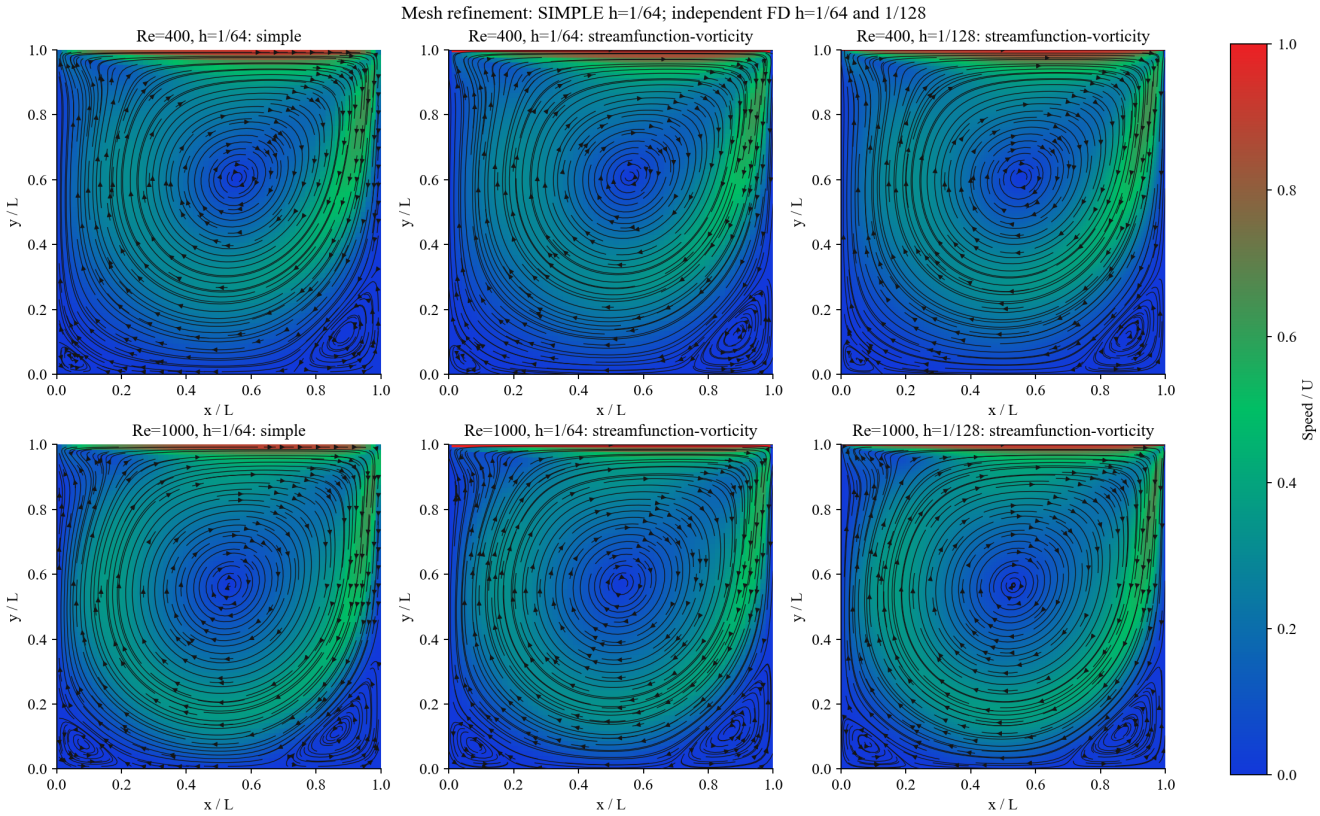
The Re=400 all-point v error retains the isolated published table entry. A separately labeled sensitivity omits that one value without replacing it. FD and FV residual magnitudes must not be compared as the same error.

Resolution can matter more than the coupling label



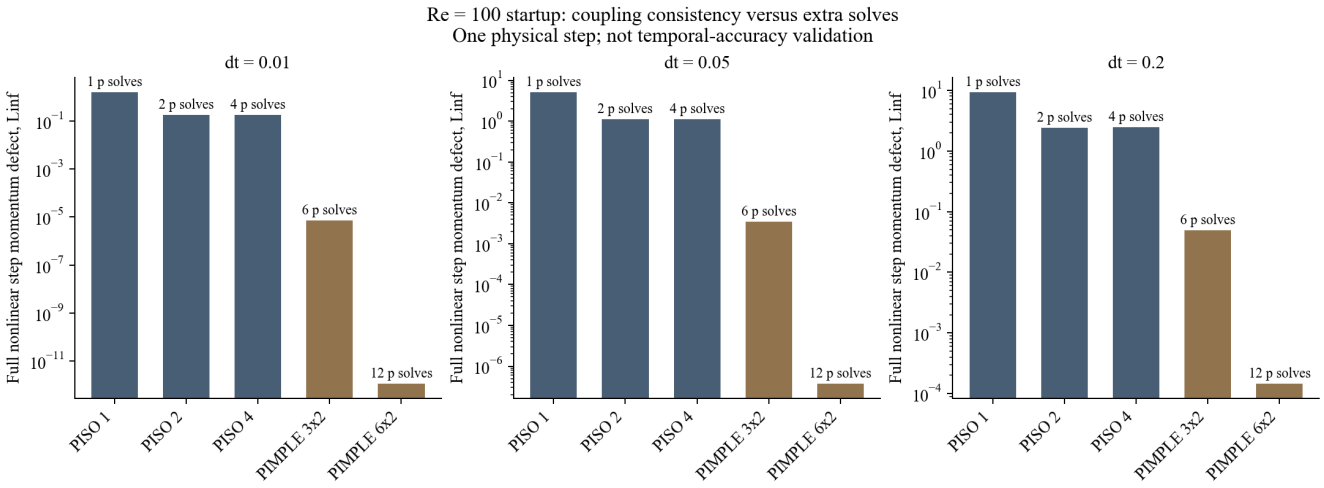
The coarse FD field may satisfy its discrete steady equation and still be inaccurate. Refining FD from h=1/32 through 1/64 to 1/128 substantially changes the velocity profiles. At Re=1000 the 128-cell FD errors are about 2.0% and 1.7%. These are velocity-reference comparisons, not a pressure validation. Retained first attempts document a failed steady gate and an exhausted step budget.

Recirculation and mesh refinement



Each panel states its mesh spacing. The sampled primary vortex and resolved corner recirculation become more reliable under refinement. Streamlines follow the computed velocity; they are not pressure or vorticity isolines. Finer or higher-Re cases can take longer and require more memory. Begin with the coarse case before selecting a laptop or Colab refinement run.

11. A measured within-step PIMPLE advantage



Fifteen configurations start from rest at Re=100 on 32 cells per side. Each takes one backward-Euler step. The full nonlinear step defect rebuilds convection from the final velocity, unlike the frozen PISO predictor defect. Each panel has its own logarithmic y range.

dt=0.05 configuration	Nonlinear Linf	Pressure solves	Momentum solves
PISO 1 inner	5.128e+00	1	2
PISO 2 inner	1.110e+00	2	2
PISO 4 inner	1.123e+00	4	2
PIMPLE 3 x 2	3.434e-03	6	6
PIMPLE 6 x 2	3.750e-07	12	12

All fifteen cell-divergence defects are below 10^{-9}. Extra inner corrections can plateau in the nonlinear defect because the predictor coefficients remain frozen. PIMPLE updates those coefficients, at higher solve cost.

What this establishes

A demonstrated reduction in within-step algebraic error. It is not a physical time-error estimate, proof of an accurate larger dt, or a universal wall-time advantage.

12. Classroom experiment and assessment plan

A. Already executed: common steady state

Re=100, 32/64 cells. Show that all three FV methods reach the same discrete steady field. SIMPLE is cheaper on the coarse grid; PISO is cheaper with the selected fine-grid settings. Keep the nonmonotone FV Ghia errors visible.

B. Already executed: one-step consistency

Re=100, 32 cells; vary inner and outer counts at each fixed dt. Compare the full nonlinear momentum defect, frozen predictor defect, mass balance and solve counts. More pressure corrections and more coefficient updates address different errors.

C. Proposed next: temporal accuracy at Re=100

Compare startup at matched final times; halve dt. Then add a smooth ramped/oscillating lid and a verified temporal reference. Compare field error, energy, amplitude/phase and cost. Use the same time scheme. Transient SIMPLE needs an outer solve inside every physical step; the current steady iteration is not a trajectory.

D. Proposed later: higher-Re qualification

Test Re=400, then 1000 with refined 64/128/256 meshes and independently validated convection. Inspect secondary vortices, centerlines and lid-wall stress. Separate spatial errors from coupling errors and physical unsteadiness.

Assess cost versus error, not a preferred winner. Keep the mesh, linear solver, initial fields and spatial equations fixed for a coupling comparison. Repeat wall-time measurements and retain failed configurations. A dt-refined numerical reference has its own error; document its refinement.

13. Exercises and the next experiment

1. Sum cell divergence and cancel shared interior face terms.
2. Add a pressure constant; verify unchanged velocity.
3. Compare one and two PISO corrections using the one-step momentum defect.
4. Check PIMPLE with one outer loop against PISO.
5. Change SIMPLE relaxation at fixed final residual tolerance.
6. Change only convection interpolation; separate spatial error from coupling effects.
7. Design a Δt-halving startup experiment before claiming a PIMPLE speedup.

Next qualification

Start with a ramped or oscillating lid. Use the same temporal scheme and matched final times. Refine Δt; measure field and phase errors. Any transient streamfunction pressure reconstruction must include acceleration terms.

Only Re=100 steady behavior is qualified in this release. Higher-Re behavior, transient accuracy and stretched grids remain separate tasks.

14. Reading, attribution and reproducibility

PySIMPLE: educational MAC finite-volume SIMPLE

NIST FiPy: pressure-correction derivation (Stokes example)

OpenFOAM: solution and algorithm control

pyOpenFOAM: another inspectable Python loop organization

Ghia, Ghia and Shin (1982): benchmark tables

This independently written module uses original implementation and explanations. External examples inform derivation and organization; their validation claims are not inherited. The audited CFD-Python PISO-titled notebook had a commented-out second correction, illustrating why source inspection matters.

Reproduce and audit

qa/run_week01_2.py executes the steady cases; qa/week01_2_teaching.py executes the one-step sweep and draws flowcharts; qa/build_week01_2.py builds the figures and notebook. tests/test_pressure_velocity.py checks conservation, pressure gauge, PISO behavior, PIMPLE limits and the common steady solution. The expanded teaching supplement provides full source links and study boundaries.

The numerical reference data, solver settings, retained failures and reproduction commands accompany the notebook and report.

Worked example: a conservative two-cell correction

Use two unit-volume cells, impermeable external faces and one shared face velocity q, positive to the right. Unit spacing and d=1 isolate the signs from mesh factors.

$$D_1 = 1, \quad D_2 = -1, \quad G_1 = -1, \quad G_2 = 1$$

For q*=1 the two cell defects are +1 and -1. Their sum vanishes because the shared face enters with opposite signs; neither cell yet satisfies continuity.

$$L = -DdG, \qquad Lp' = -Dq^*$$

Here L has rows (1,-1) and (-1,1). Pin p2'=0. The first row gives p1'=-1; then Gp'=1 and q_new=1-1=0. Adding the same constant to both pressures changes no face gradient. This is the constant pressure nullspace.

Exercise with solution

Change q*=2 and d=0.5. L becomes half the previous matrix. With p2'=0, p1'=-4, Gp'=4 and q_new=2-0.5(4)=0. Applying only alpha_p of the face correction would leave a cell mass defect. In SIMPLE, alpha_p relaxes the pressure update, while the velocity correction remains conservative.

This is a nondimensional sign-and-conservation control, not a resolved cavity. Before using the sparse solver, write the same cancellation for three cells and verify D times a constant internal through-flux at the impermeable ends.